

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 14-2023: A 31-Year-Old Man with Redness of the Right Eye

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PRESENTATION OF CASE

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Dr. Aaron R. Kaufman (Ophthalmology, Massachusetts Eye and Ear): A 31-year-old man was evaluated in the emergency department of this hospital because of redness of the right eye.

The patient had been in his usual state of health until 8 days before the current evaluation, when he noticed an itchy perianal rash and rectal pain. After 2 days, the symptoms did not abate, and he sought evaluation in the urgent care clinic of this hospital.

On evaluation, the patient reported the perianal rash and gradually increasing rectal pain. He had been sexually active with two male partners in the previous month. On examination, the temporal temperature was 36.7°C, the blood pressure 117/84 mm Hg, and the pulse 80 beats per minute. Three scattered, tender, shallow ulcers and a vesicle were present in the perianal area. No rectal discharge or fissures were seen. Tender inguinal lymphadenopathy was present bilaterally; there was no cervical or axillary lymphadenopathy. Tests for human immunodeficiency virus (HIV) infection, gonorrhea, chlamydia, syphilis, and orthopoxvirus infection were performed. Empirical treatment with valacyclovir was started. The patient was discharged home with instructions to await the test results.

Two days before the current evaluation, test results were available. The test for HIV types 1 and 2 was nonreactive, as were the tests of urinary and rectal specimens for *Neisseria gonorrhoeae* and *Chlamydia trachomatis* nucleic acids. The blood test for treponemal antibodies was positive; the results of rapid plasma reagin (RPR) testing were not yet available. Polymerase-chain-reaction (PCR) testing of a swab specimen obtained from the perianal vesicle detected the presence of nonvariola orthopoxvirus DNA. Species-specific real-time PCR testing confirmed the presence of mpox virus (MPXV; formerly known as monkeypox virus), clade II. The patient was called and instructed to return to the emergency department of this hospital.

On evaluation, the patient reported a new rash on his chest, abdomen, and back that was similar to the perianal rash. He reported new fever, chills, fatigue, malaise,

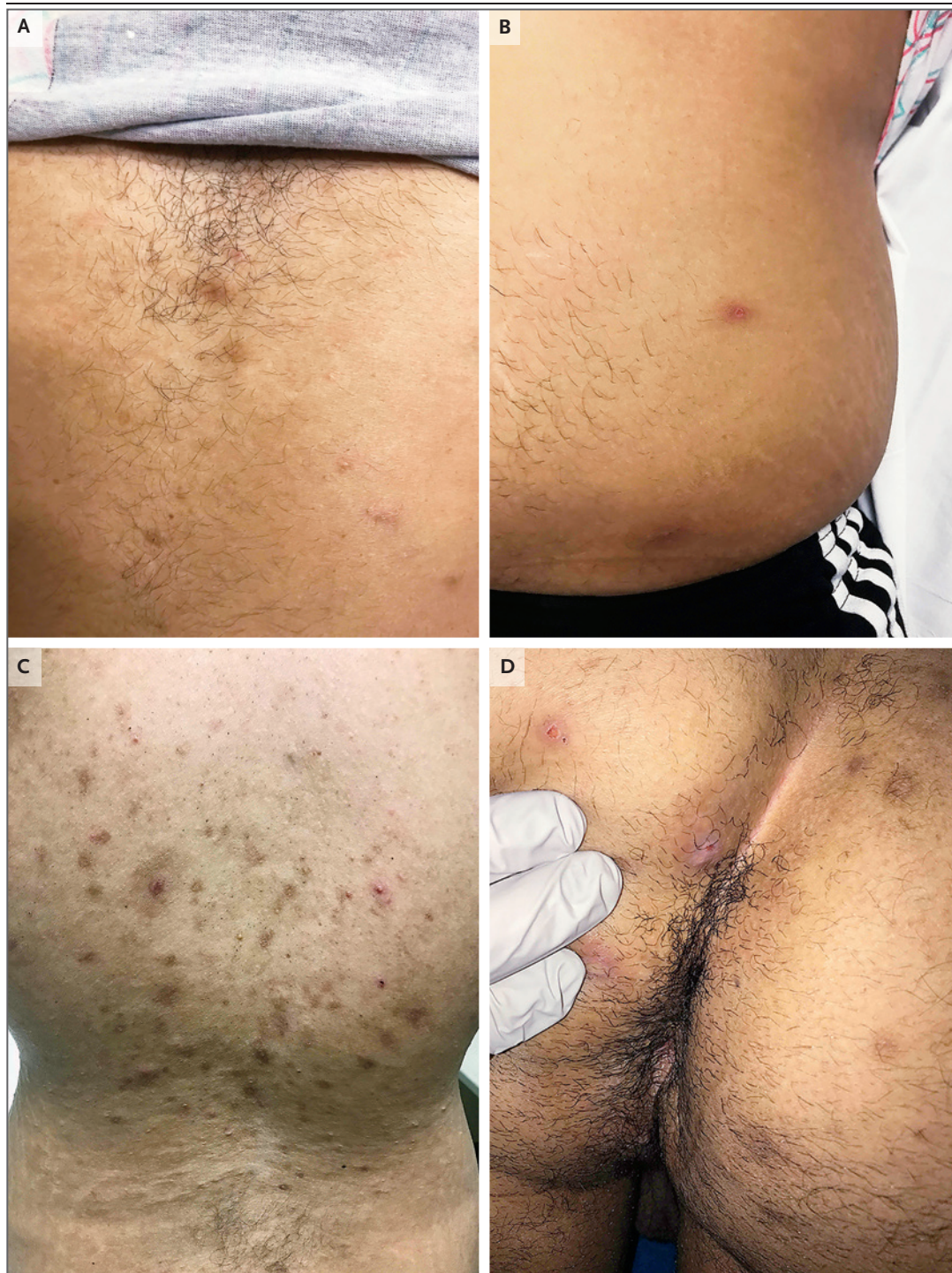


Figure 1. Photographs of the Skin.

Photographs of the skin were obtained 2 days before the current presentation. Scattered, tender, shallow ulcers are present on the chest (Panel A), abdomen (Panel B), and back (Panel C). Several tender, umbilicated pustules and a tender, shallow ulcer are present in the perianal area (Panel D).

and headache. There was no weight loss, sore throat, or cough. Numerous scattered, tender, shallow ulcers were present on the chest, abdomen, and back, in addition to five tender, umbilicated pustules and one shallow ulcer in the perianal area (Fig. 1). Treatment with tecovirimat was started. The patient was discharged home with instructions to follow up in the sexual health clinic of this hospital.

During the next 2 days, the patient took tecovirimat. On the day of the current evaluation, he called the sexual health clinic because of redness and throbbing, pressurelike pain of the right eye. He was instructed to return to the emergency department of this hospital.

On the current evaluation, the patient reported that redness and discomfort of the right eye had started after he had been discharged from the emergency department and had increased in severity and become painful during the subsequent 2 days. He rated the pain at 10 on a scale of 0 to 10, with 10 indicating the most severe pain. He did not have blurry vision, photophobia, flashing lights, or floaters. There were no symptoms in the left eye. Since the initiation of antiviral therapy, the fever, chills, and headache had resolved, and the rectal pain had decreased in severity. There was no rhinorrhea or cough.

The patient had a history of syphilis, which had occurred 5 years earlier and had been treated with intramuscular penicillin G. Since receiving the diagnosis of mpox, he had learned that syphilis and mpox had been diagnosed in one of his sexual partners from the previous month. Medications included tecovirimat and valacyclovir. There were no known drug allergies. The patient lived alone in a suburban area of New England and worked in the food-service industry. He did not smoke cigarettes, drink alcohol, or use illicit drugs. He did not know his family history.

On examination, the temporal temperature was 36.4°C. The uncorrected visual acuity was 20/20 in each eye. The pupils were symmetric and reactive to light, without a relative afferent pupillary defect, and extraocular movements were full. The intraocular pressure was 16 mm Hg in the right eye and 17 mm Hg in the left eye (normal range, 10 to 21). When proparacaine drops were administered to check the intraocular pres-

sure, the pain in the right eye did not decrease in severity. There was mild tenderness on palpation of the right globe under the closed eyelid. Slit-lamp examination revealed mild meibomian-gland dysfunction in both eyes; there was no eyelid edema or erythema. The right conjunctiva was diffusely injected, with severe bulbar hyperemia (Fig. 2A). The right caruncle was boggy and erythematous. On eversion of the eyelids, small-to-medium-sized follicles and tarsal conjunctival petechiae were observed on the right lower and upper lids (Fig. 2B and 2C). The left conjunctiva was normal. The cornea, anterior chamber, iris, and lens appeared normal in both eyes. Dilated funduscopic examination was performed. The vitreous, optic nerve, retinal vasculature, macula, and periphery appeared normal in both eyes.

A diagnosis was made.

DIFFERENTIAL DIAGNOSIS

Dr. Miriam B. Barshak: Severely painful, unilateral acute conjunctivitis with small-to-medium-sized follicles developed in this patient while he was taking tecovirimat for mpox. Severe pain is unusual in patients with conjunctivitis and suggests the possibility of scleritis, keratitis, or uveitis, but a detailed examination revealed no involvement of other anatomical regions of the eye. Follicles are associated with a lymphocytic inflammatory response, which is often seen in patients with conjunctivitis caused by a viral infection or chlamydia.

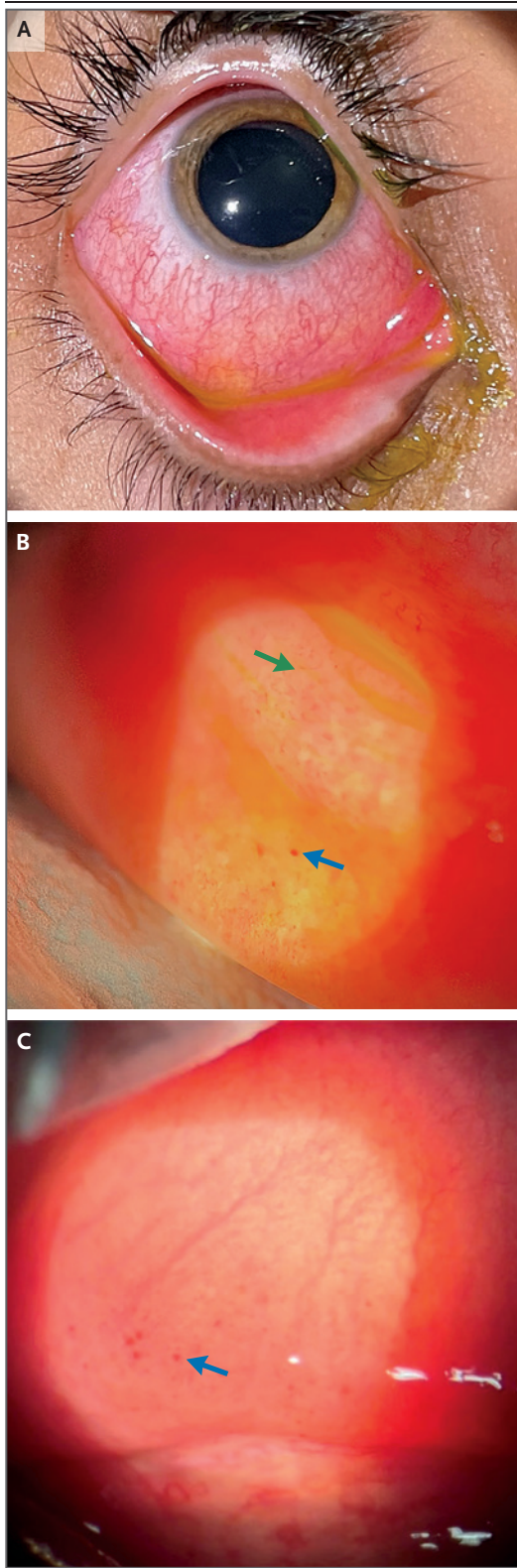
Acute conjunctivitis is the most common cause of eye redness that leads to medical visits, affecting an estimated 6 million people annually in the United States and accounting for approximately 1% of all primary care visits and 30% of all eye-related emergency department visits.¹ In constructing a differential diagnosis, I will consider the noninfectious and infectious causes of acute conjunctivitis, with a particular focus on oculogenital syndromes.

NONINFECTIOUS CAUSES OF ACUTE CONJUNCTIVITIS

Allergic conjunctivitis is the most common cause of acute conjunctivitis, affecting 15 to 40% of the population. It is associated with itch-

Figure 2. Photographs of the Right Eye.

Photographs of the right eye were obtained on the current presentation. On examination of the right eye (Panel A), diffuse bulbar and tarsal conjunctival hyperemia is present. With pharmacologic dilation of the pupil and the administration of fluorescein dye, there is no staining of the ocular surface. On eversion of the right lower and upper eyelids (Panels B and C, respectively), tarsal conjunctival and subconjunctival petechial hemorrhages (blue arrows) and small-to-medium-sized conjunctival follicles (green arrow) are visible.



ing in both eyes, sneezing, and rhinorrhea, but unilateral involvement and severe pain would be very atypical for allergic conjunctivitis.¹ Another noninfectious cause is toxic conjunctivitis, which is usually induced by topical ocular medications. Ocular side effects have not been described in association with oral tecovirimat.^{2,3} Although allergic, toxic, and other types of conjunctivitis need to be considered, infections are the most common cause of medical visits for allergic conjunctivitis.¹

BACTERIAL INFECTIONS

Bacterial infections cause less than 20% of cases of acute conjunctivitis in adults. Purulent drainage is suggestive of acute bacterial conjunctivitis, but this finding has poor sensitivity and specificity for the diagnosis. The combination of crusting of the eyelashes in both eyes, the absence of itching, and the absence of previous episodes of conjunctivitis has a better positive predictive value.^{4,5} This patient had no history or examination findings of drainage, so acute bacterial conjunctivitis is unlikely. However, I will consider some types of acute bacterial conjunctivitis when reviewing oculogenital syndromes.

VIRAL INFECTIONS

Viral infections cause 75 to 80% of cases of acute conjunctivitis in adults. Watery drainage is a classic sign of acute viral conjunctivitis, but the combination of concurrent pharyngitis, preauricular lymphadenopathy, and recent contact with another person with eye redness has a better positive predictive value. Overall, 65 to 90% of cases of acute viral conjunctivitis result from adenovirus infection, which is associated with pharyngoconjunctival fever and epidemic kerato-

conjunctivitis, both of which are associated with periauricular lymphadenopathy. This patient did not have these associated findings, so the likelihood of acute adenoviral conjunctivitis is low.¹

Herpes simplex virus (HSV) and varicella-zoster virus (VZV) infections account for less than 10% of cases of acute conjunctivitis. These viral infections typically cause unilateral symptoms and may be associated with skin lesions on the eyelid, keratitis, or uveitis.¹ Acute VZV conjunctivitis is usually associated with infection of the ophthalmic branch (V_1) or maxillary branch (V_2) of the trigeminal nerve. HSV and VZV infections typically abate with valacyclovir therapy, so acute HSV or VZV conjunctivitis is unlikely in this patient.

Enterovirus infection, measles, mumps, rubella, severe acute respiratory syndrome coronavirus 2 infection, Ebola virus disease, mpox, and other viral infections can also cause acute conjunctivitis. To narrow the differential diagnosis in this case, I will integrate the patient's history of sexually transmitted infections (STIs).

OCULOGENITAL SYNDROMES

Many STIs can cause ocular disease. It is therefore imperative to elicit a thorough history from patients with ocular symptoms to assess the risk of STI acquisition, identify possible symptoms of STIs, and review the diagnosis and treatment of past and current STIs, while recognizing that urogenital testing may not rule out ocular infection. This patient had received intramuscular treatment for syphilis 5 years earlier, was currently being treated for mpox, and had recently had negative testing for other STIs, including HIV infection and urogenital gonorrhea and chlamydia.

GONORRHEA AND CHLAMYDIA

N. gonorrhoeae infection is the most severe oculo-genital infection that can cause acute conjunctivitis, with a hyperacute onset of copious purulent drainage, eyelid swelling, eye tenderness, preauricular lymphadenopathy, and decreased vision. There is a high risk of corneal involvement, including perforation. *C. trachomatis* infection classically has unilateral involvement and a subacute onset; mucopurulent drainage occurs over a period of weeks to months, and a tender preauricular lymph node and follicles are usually detected on eversion of the eyelid. This pa-

tient's syndrome is not typical for acute gonococcal or chlamydial conjunctivitis.¹

MPOX

Mpox is an important consideration in this patient. The worldwide mpox outbreak in 2022 and 2023 has been associated with sexual contact. In the literature that predated the current outbreak, ocular disease was a frequent complication. In a 2014 report of cases occurring in the Democratic Republic of Congo, 23% of patients with confirmed mpox had ocular manifestations, including conjunctivitis (in approximately 20% of these patients), blepharitis (in approximately 30%), severe keratitis (in approximately 7.5%), photophobia (in approximately 22%), and decreased visual acuity (in 5 to 10%). Most of these patients were children younger than 10 years of age who had severe disease.^{6,7}

During the current outbreak, the following ocular manifestations have been described: eyelid pox lesions, conjunctival involvement (follicles, papules, vesicles, ulcers, or pseudomembranes), subconjunctival nodules, blepharitis, and keratitis. However, ocular involvement has been much less common during the current outbreak than during previous outbreaks, affecting less than 1% of patients in retrospective case reporting, even before the use of antiviral treatment.⁸⁻¹⁰ Several cases of mpox conjunctivitis have been described in which symptoms developed or worsened while the patient was taking tecovirimat.¹¹

SYPHILIS

Any discussion of oculogenital syndromes must include syphilis. The incidence of ocular syphilis has risen in the United States in the past 10 years,¹² mirroring the increased incidences of primary and secondary syphilis. Ocular symptoms are most likely to occur with secondary syphilis, typically 2 to 6 months after infection, but can also occur with tertiary syphilis and rarely with primary syphilis. In patients with secondary syphilis, ocular symptoms often have an acute onset, whereas in patients with tertiary syphilis, such symptoms may develop insidiously for 1 to 2 months before the patient seeks care. Ocular syphilis may occur without other signs or symptoms of syphilis. It is important to note that ocular syphilis may occur after primary or secondary syphilis has been treated with weekly

intramuscular penicillin, with or without oral doxycycline, because these agents do not reach therapeutic drug levels in the eye.

Syphilis can affect any region of the eye. Uveitis is the most common ocular manifestation, occurring in 75 to 90% of cases of ocular syphilis. Conjunctival involvement can include a chancre (primary disease), papillary conjunctivitis (secondary disease), or granulomatous or gummatous conjunctivitis (tertiary disease).^{13,14} This patient did not have any of these disease manifestations, and isolated conjunctivitis would be very unusual for ocular syphilis.¹⁵

Could this patient have both mpox and syphilis? Both diseases are associated with sexual contact, and several cases of coinfection have been reported in the literature.^{16,17} However, the timing of the onset of this patient's ocular disease — relative to either his history of syphilis (treated 5 years earlier) or a possible new case of syphilis (related to new exposure in the previous month) — would be highly unusual, as would the isolated conjunctivitis without a conjunctival chancre or papillary, granulomatous, or gummatous findings. Therefore, I do not think that syphilis is the cause of his ocular syndrome.

Nevertheless, syphilis is both a vision-threatening and treatable disease, and it is critical to rule out the diagnosis with a nontreponemal test (RPR test). In patients with a history of syphilis, treponemal antibodies persist regardless of the efficacy of previous treatment. New or active syphilis would be suggested by an RPR titer that is at least 4 times as high as the most recently obtained (post-treatment) RPR titer.

I suspect that this patient's RPR test was negative, which would leave mpox conjunctivitis as the most likely diagnosis. To establish this diagnosis, I would obtain a conjunctival swab or biopsy specimen to be tested for mpox nucleic acids.

DR. MIRIAM B. BARSHAK'S DIAGNOSIS

Mpox conjunctivitis.

DIAGNOSTIC EXAMINATION

Dr. Roberto Pineda: The patient's ophthalmic examination (Fig. 2) revealed findings consistent with unilateral acute conjunctivitis. He had a

history of syphilis but did not have specific ocular manifestations of syphilis, such as an eyelid or conjunctival chancre or a conjunctival gumma, on examination. Although these manifestations have been described, they are rare. In patients with secondary syphilis, mild conjunctivitis is common, but systemic symptoms are typically much more prominent than they were in this patient.¹⁸

Acute viral conjunctivitis is typically a clinical diagnosis. Ancillary testing is usually not indicated but may support the diagnosis of some conditions. Mpox conjunctivitis, for example, is often a clinical diagnosis that is supported by the results of nucleic acid amplification testing (NAAT) of specimens obtained from nonocular sites.¹⁰ However, since the time of this patient's evaluation, the Centers for Disease Control and Prevention (CDC) has approved the use of a conjunctival swab in NAAT for mpox.¹⁹

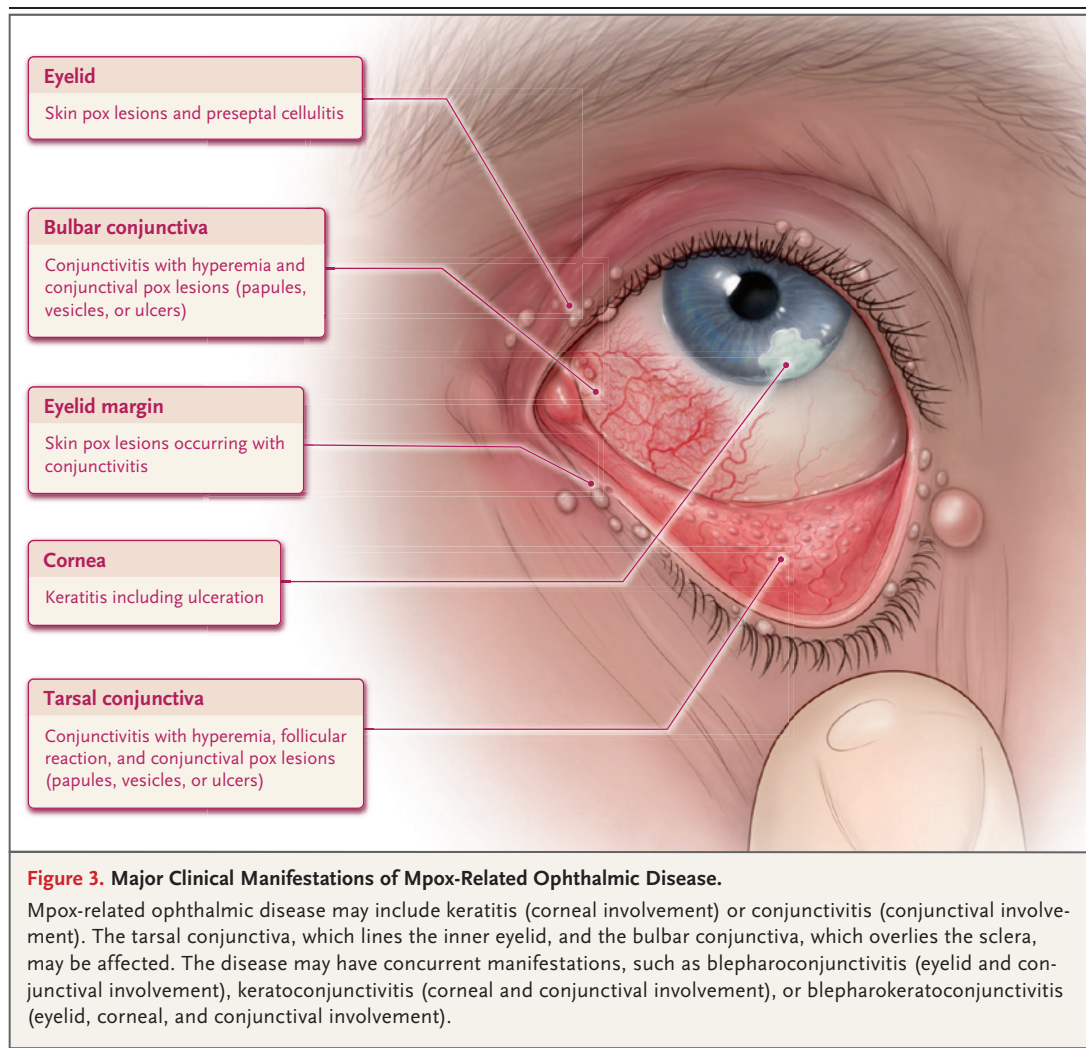
CLINICAL DIAGNOSIS

Mpox conjunctivitis.

MPOX-RELATED OPHTHALMIC DISEASE

Dr. Pineda: The understanding of mpox-related ophthalmic disease (MPXROD) in the current outbreak of clade IIB MPXV comes from reports describing the 2022 and 2023 global outbreak,^{10,11} literature regarding clade I MPXV,^{20,21} and literature regarding the variola (smallpox) and vaccinia viruses.²² Manifestations of MPXROD (Fig. 3) may include skin lesions on the eyelid, conjunctivitis, keratitis, or a combination of these conditions.¹⁰

MPXROD appears to be less common in the current outbreak than in previous outbreaks.¹⁰ Eyelid involvement has not been reported separately from other facial involvement in the current outbreak, so the precise incidence is not known. Conjunctivitis has occurred in less than 1% of cases in the current outbreak,⁹ as compared with 7 to 30% of cases in previous outbreaks of clade I MPXV.²⁰ Keratitis has occurred rarely in the current outbreak²³ but was reported to occur in 1 to 4% of cases in previous outbreaks of clade I MPXV.²¹ During previous clade I MPXV outbreaks, MPXROD was less likely to occur in vaccinated persons than in unvaccinat-



ed persons^{20,21}; whether this has been true during the current outbreak is not known. Previous clade I MPXV outbreaks were associated not only with involvement of the cornea by the mpox viral process but also with bacterial superinfection of the cornea.²⁴

In a patient with mpox or exposure to mpox, MPXROD is likely to result from autoinoculation (i.e., eye rubbing).^{11,25} MPXROD usually occurs in patients who also have nonocular lesions,²⁶ which may be painful or uncomfortable.^{21,24} Touching these lesions and then rubbing the eye may inoculate the virus into the ocular surface. In addition, the presence of eyelid lesions might lead to inoculation of the conjunctiva; concurrence of lesions on the eyelid margin and conjunctivitis has been reported.^{21,25} Confirmed positive results of NAAT and viral culture of a

swab specimen obtained from the ocular surface have been reported in patients with mpox conjunctivitis.^{11,25} Infection of one eye might lead to inoculation of the other eye. Variation in eye-rubbing behavior among patient populations might explain differences in the incidence of MPXROD between the current and previous outbreaks. In previous outbreaks, children were predominantly affected,^{20,21} and children are more likely to rub their eyes than adults; consequently, autoinoculation of the virus into the eyes may have been more common. In the current outbreak, children have been affected only rarely.

DISCUSSION OF MANAGEMENT

Dr. Caitlin M. Dugdale: Although the cutaneous manifestations of mpox are increasingly well

recognized, several extracutaneous manifestations of mpox, including MPXROD, can develop and become severe (Fig. 4). In case series and cohort studies describing the current mpox outbreak, 6 to 36% of patients have reported rectal pain, rectal bleeding, or other symptoms of proctitis, and 10 to 21% of patients have had pharyngitis, epiglottitis, or ulcerative tonsillitis.^{26,29,32,34-37} Painful lymphadenopathy, particularly in the cervical or inguinal region, has affected 41 to 85% of patients in the current outbreak.^{26,29,34-37} Mpox-related encephalitis, encephalomyelitis, myocarditis, and pneumonia have all been reported, but these manifestations are rare.^{26,27,30,31}

Although tecovirimat has not yet been approved by the Food and Drug Administration (FDA) for the treatment of mpox, it is the most commonly used antiviral therapy for this condition.⁹ Tecovirimat was first developed as a treatment for smallpox, and it received FDA approval for that indication on the basis of data from studies in animals (i.e., studies of MPXV infection in nonhuman primates and rabbitpox virus infection in rabbits) because studies in humans were not feasible.³⁸ Tecovirimat is currently available for the treatment of mpox only as an investigational drug, and its efficacy against mpox has not yet been established in clinical trials in humans, although such trials are under way.³⁸⁻⁴⁰

It is notable that this patient's ocular symptoms evolved while he was taking tecovirimat, despite the rapid resolution of his skin lesions after initiation of that therapy. Data regarding tecovirimat levels in the surface or deeper structures of the eye are lacking.⁹ However, studies in animals suggest that tecovirimat penetrates the brain and cerebrospinal fluid.⁴¹ Emerging data suggest that tecovirimat has a low barrier to resistance, given that it works by inhibiting the VP37 viral protein shared by smallpox viruses and MPXV; even a single amino acid substitution in VP37 can lead to a reduction in antiviral activity. Tecovirimat resistance was thought to be unlikely in this case because the patient's skin lesions had completely resolved 2 weeks after initiation of treatment; in addition, delayed resolution of ocular symptoms in mpox has been reported.^{11,42} Given the low barrier to resistance and unknown efficacy of tecovirimat, the CDC currently recommends the use of tecovirimat only in patients with severe disease, a risk of

severe disease, or involvement of vulnerable anatomical areas.

Dr. Pineda: The eye is considered to be a vulnerable anatomical site and is susceptible to serious consequences of infection, such as scarring or severe pain. Therefore, most patients with MPXROD are treated with systemic antiviral therapy.¹⁹

Supportive topical agents, including ophthalmic lubricants and prophylactic antibiotic drops, are often provided for MPXROD and were given to this patient.²⁴ A role for topical trifluridine in the treatment of MPXROD has been suggested; the CDC currently states that its use "may be considered."^{11,19} Trifluridine has been approved by the FDA for the treatment of HSV infection but has known corneal toxic effects, such as epithelial injury and corneal neurotrophía.¹⁰ In vitro studies have shown efficacy of trifluridine against orthopoxvirus infection and ocular vaccinia infection,²² but published cases describing the use of trifluridine for the treatment of mpox are limited.¹¹

If there is corneal involvement, the course of MPXROD may be prolonged.¹¹ However, if there is conjunctivitis without corneal involvement, the prognosis is better, with the disease usually resolving within several weeks.^{11,42} Resolution of MPXROD may lag behind resolution of nonocular disease.⁴² The reason for this difference is currently unclear, although variation in viral tropism, ocular immune privilege, decreased penetration of tecovirimat in eye structures, or autoinoculation of the virus into the eye several days after the onset of cutaneous mpox may have a role.

Infection control is an essential component of mpox management and was prioritized in this case.¹⁰ Infection-control measures include avoidance of moving the patient beyond the examination room or within the facility¹⁹; use of personal protective equipment, such as an N95 respirator, eye protection, a gown, and gloves¹⁹; and decontamination of surfaces and minimal use of shared equipment that is not disposable or easily disinfected.¹⁹ It is also appropriate to consider the selective use of telehealth for mild or resolving cases of MPXROD to minimize potential spread in health care settings.

It is important that patients with MPXROD remain at home until the quarantine period has ended, in accordance with guidance from public

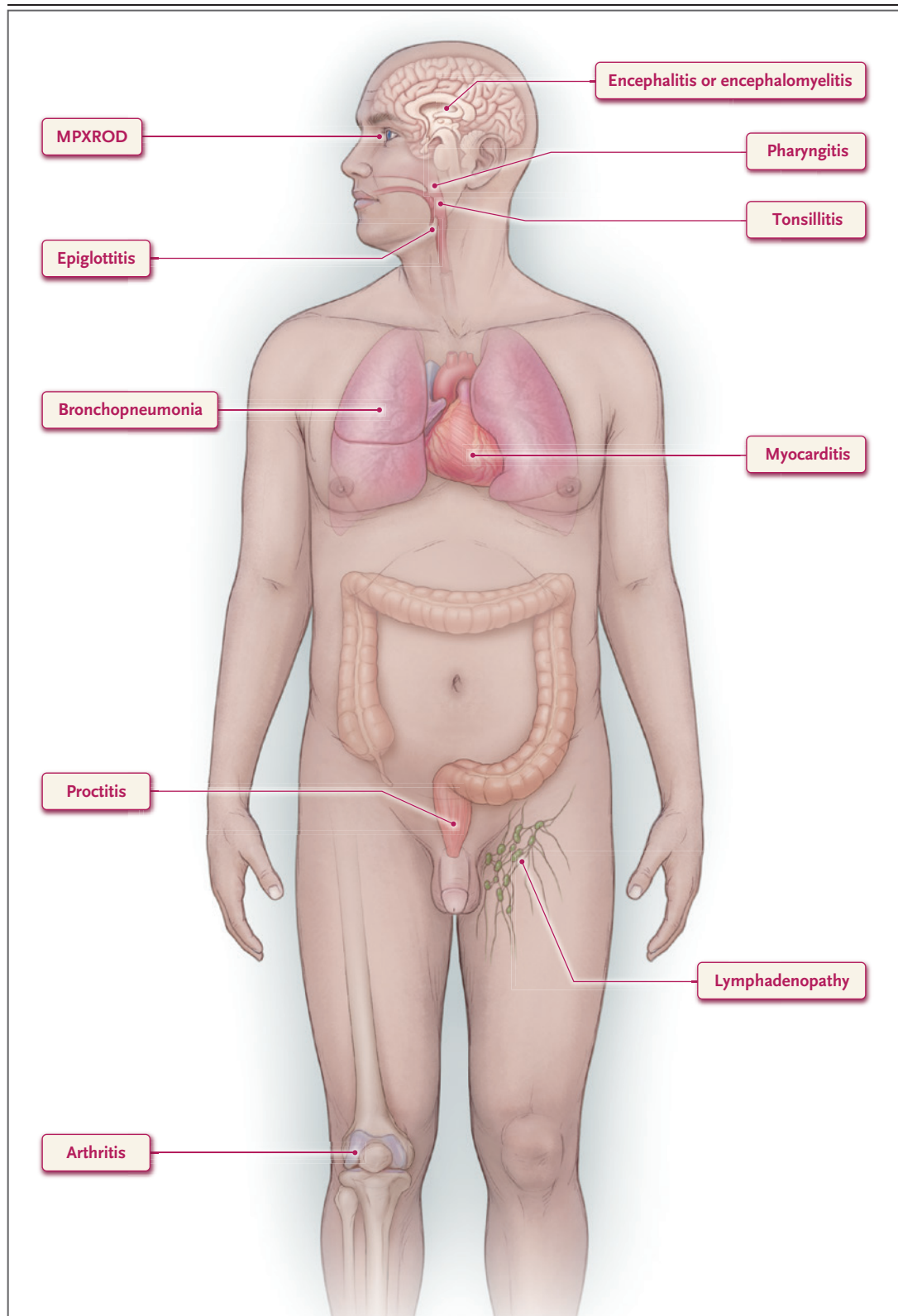


Figure 4 (facing page). Extracutaneous Manifestations of Mpox.

Numerous extracutaneous manifestations of mpox have been reported: encephalitis or encephalomyelitis^{27,28}; mpox-related ophthalmic disease (MPXROD), including skin lesions on the eyelid, conjunctivitis, and keratitis^{10,11}; pharyngitis²⁶; tonsillitis²⁹; epiglottitis²⁶; bronchopneumonia³⁰; myocarditis^{26,31}; proctitis^{26,32}; lymphadenopathy^{26,29}; and arthritis.³³

health authorities.¹⁹ The use of strict hand hygiene and the avoidance of eye rubbing are essential to prevent autoinoculation.^{11,19} Patients are advised to avoid wearing contact lenses until disease resolution¹¹ and to disinfect contaminated surfaces.¹⁹

FOLLOW-UP

Dr. Kaufman: After the clinical diagnosis of mpox conjunctivitis was made, the patient had video follow-up visits every 3 days. He continued taking tecovirimat twice daily for 2 weeks and used artificial tears 5 times daily to lubricate the ocular surface. Treatment with trifluridine was not started because there was no corneal involvement. Quarantine at home was instructed by the Massachusetts Department of Public Health.

While awaiting the RPR test results, the patient received empirical treatment for syphilis of unknown duration with a dose of intramuscular penicillin G. Ultimately, the RPR test was non-reactive.

By the first video follow-up visit, the pain in the right eye had resolved, and the eye redness had decreased. However, new pain and redness had developed in the left eye; these findings were consistent with bilateral sequential mpox conjunctivitis that was most likely due to autoinoculation. After the patient had taken tecovirimat for 2 weeks, the cutaneous mpox lesions had resolved, and the eye redness had decreased. Massachusetts Department of Public Health guidance cautions that patients may remain contagious until all ocular symptoms have resolved. Therefore, tecovirimat was provided for an additional 2 weeks. When the eye redness resolved, the patient was released from home quarantine.

FINAL DIAGNOSIS

Mpox conjunctivitis.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

We thank Dr. Aaron R. Kaufman for the clinical photographs and for his assistance with the case presentation.

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